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WORK EXPERIENCE

- **September 2025 – Present:** Postdoctoral Researcher, IRC for Smart Mobility and Logistics, King Fahd University of Petroleum & Minerals (KFUPM), Dhahran, Saudi Arabia.
- **April 2025 – Present:** Visiting Researcher at the University of the Ryukyus, Japan and **Assistant Professor** at Zagazig University, Egypt.
- **October 2022 – March 2025:** PhD Student at Interdisciplinary Intelligent Systems Engineering Department, University of the Ryukyus, Japan
- **July 2019 – October 2022:** Teaching Assistant at Electrical Power and Machines Engineering Department - Zagazig University.
- **October 2016 – July 2019:** Demonstrator at Electrical Power and Machines Engineering Department - Zagazig University.

EDUCATION

- **October 2022 – March 2025:** Ph.D. in Integrated Intelligence Engineering, University of the Ryukyus, Japan
 - **Ph.D. Title:** Optimal operation and management of hybrid renewable energy systems
 - **Grade:** Outstanding, CGPA: 4.0/4.0
- **October 2016 – July 2019:** Master (M.Sc.) in Electrical Power and Machines Engineering, Zagazig University, Zagazig, Egypt.
 - **M.Sc. Title:** Operating Reserve Determination in Renewable Power Integration Studies
 - **Preliminary Courses:** 3.3 of 4 GPA.
 - **Graduate Courses:** Power System Quality, Induction Machine, Economic Power System, HV & LV Power System Protection, Numerical Analysis, Electronic Measurement.
- **Sept 2011 - Jul 2016:** B.Sc. in Electrical Power and Machines Engineering, Zagazig University, Zagazig, Egypt.
- **Final Grade:** Excellent with honors, 90.01%.
- **Graduation Project:** Electrical Installation For Tourist Village.
- **Graduation Project Grade:** Excellent

Research Area

My research and work interests concentrate but are not limited to the following topics: **Power System Protection and Control, Grid Stability and Power Quality, Smart Grids, Renewable energy integration, Computer programs using simulation models and Artificial intelligence, Multi-Objective Optimization, Energy Management, EVs Technology (V2X).**

Languages

- **Arabic (Native Language)** Fluent in writing and speaking
- **English (Foreign Language)** Very good in writing and speaking

Achievements

- **Awarded the President's Honorary Award**, University of the Ryukyus, for outstanding performance in the Doctor of Engineering Program, which is the **university's highest recognition for a graduate student**. Distinguished by publishing **11 high-impact journal papers** in prestigious platforms such as **Elsevier and IEEE**, showcasing cutting-edge research **during his Ph.D.**
- Received recognition for developing innovative real-time management systems for microgrids and autonomous energy systems, combining theoretical research and practical applications.
- Awarded the **Zagazig University Award** for International Publications in 2021, 2022, and 2023 for significant contributions to globally recognized research.
- **Awarded the Monbukagakusho Scholarship**, Government of Japan, in October 2022 for the Special Doctoral Program in Resilient Smart City Program under the International Priority Graduate Program (PGP) from the Ministry of Education, Culture, Sports, Science, and Technology, Japan.
- **Co-Principal Investigator**, multi-institutional project funded by the Research Development and Innovation Authority (SAR 4.53M) on *EV and Renewable Energy Integration for Grid Stability*, with collaborators from Saudi Arabia, Egypt, and Japan.
- **Co-Principal Investigator**, Saudi Basic Science Research Consortium Grant project on *Green Hydrogen and Smart Cities for Net-Zero 2060*, in collaboration with partners from Saudi Arabia, Egypt, Japan, and China

Administrative and Institutional Service

- Member, **Projects, Innovation, Entrepreneurship, and International Ranking Management Unit** – Faculty of Engineering, Zagazig University (Appointed by Faculty Dean's Decision No. 22, October 14, 2025).
 - Contributed to developing strategies for project management, innovation support, entrepreneurship programs, and improving the faculty's international academic ranking.
- Member, **Academic Accreditation Committee** – Faculty of Engineering, Zagazig University, supporting quality assurance documentation and compliance with national accreditation standards.
- **Member, Committee for Preparing and Updating Postgraduate Studies and Research Programs** - Faculty of Engineering, Zagazig University
 - Participated in reviewing, developing, and updating postgraduate academic programs and research regulations.
 - Contributed to aligning postgraduate curricula with National Academic Reference Standards (NARS).

Projects and Research Collaboration

- **International Research Collaboration – Jazan University, Saudi Arabia; University of the Ryukyus, Japan**
Co-Principal Investigator in a multi-institutional research project funded by the Research Development and Innovation Authority (SAR 4.53M) on Managing the Impact of Electric Vehicles and Renewable Energy Integration on Grid Stability and Power Quality.
- **International Research Collaboration – Jazan University (Saudi Arabia); Yanbu Industrial College (Saudi Arabia); Zagazig University (Egypt); University of the Ryukyus (Japan); Central South University (China)**
Co-Principal Investigator in a consortium project funded by the Saudi Basic Science Research Consortium Grant (BSRCG) on Integrating Green Hydrogen, Renewable Energy, and Intelligent Infrastructure in Smart Cities to Accelerate Saudi Arabia's Path Toward Net-Zero Emissions by 2060
- **EU-AU Long-Term Joint Research and Innovation Partnership on Sustainable Energy (Under Review)**
Submitted a joint proposal under this EU-AU initiative in collaboration with: Technical University of Berlin (Germany), Polytechnic University of Milan (Italy), Zagazig University (Egypt), and Thies Polytechnic School (Senegal)
- **International Research Collaboration – Central South University, China**
Participated in joint research activities with scholars from Central South University on topics related to sustainable energy and electrical engineering.
- **Academic Accreditation Contribution – Zagazig University**
Contributed to quality standards and documentation during the academic accreditation process of the Department of Electrical Power and Machines Engineering, Faculty of Engineering, Zagazig University.

COMPUTER AND TECHNICAL SKILLS

- MATLAB, AutoCAD, ETAP, Microsoft Office Word, Excel and PowerPoint, COMSOL Multiphysics

Master Research

- A new approach for integrating wave/PV/wind hybrid energy has been presented. The three proposed energies in this research are successfully integrated together (experimental and simulation).
- A new experimental model of wave energy has been presented using a special wave flume and flap generator.

Society Membership

- Member and Reviewer for IEEE Access journal
- Member and Reviewer for energy conversion and Management
- Member and Reviewer for Computers & Electrical Engineering
- Member and Reviewer for International Journal of Green Energy
- Member and Reviewer for Energy journal.

Internet Research

Google Scholar	https://scholar.google.com/citations?user=Y2ATJeoAAAAJ&hl=ar
ORCID	https://orcid.org/0000-0003-3615-3714
Mendeley	https://www.mendeley.com/profiles/mahmoud-elkholy6/
Research gate	https://www.researchgate.net/profile/M-Elkholy
Publons	https://publons.com/researcher/3423654/mahmoud-elkholy/
Web of Science	https://www.webofscience.com/wos/author/record/AAF-8819-2020

Publications

1. Song, Dongran, Asifa Yousaf, Javeria Noor, Yuan Cao, Mi Dong, Jian Yang, Rizk M. Rizk-Allah, **M. H. Elkholy**, and M. Talaat. "ANN-Based Model Predictive Control for Hybrid Energy Storage Systems in DC Microgrid." *Protection and Control of Modern Power Systems* 10, no. 4 (2025): 1-15.
2. A. S. Irshad, A. S. Amin, A. M. Ilham, **M. H. Elkholy**, S. Elias, and T. Senjyu, "Power capacity enhancement of hydropower plant through the penetration of solar and wind energy," *International Journal of Electrical Power & Energy Systems*, vol. 169, p. 110787, 2025/08/01/ 2025, doi: <https://doi.org/10.1016/j.ijepes.2025.110787>.
3. Ahmed, Shoaib, **M. H. Elkholy**, M. Talaat, Tomonobu Senjyu, Akie Uehara, Dongran Song, Taghreed Said, Mahmoud M. Gamil, and Mohammed Elsayed Lotfy. "Dynamic hybrid energy management for self-sufficient residential microgrids: A flexibility-constrained approach with integration of hybrid backup storage systems." *Journal of Energy Storage* 138 (2025): 118745.
4. **M. H. Elkholy**, A. Yona, S. Ueda, T. Said, T. Senjyu, and M. E. Lotfy, "Experimental Investigation of AI-Enhanced FPGA-Based Optimal Management and Control of an Isolated Microgrid," *IEEE Transactions on Transportation Electrification*, vol. 10, no. 2, pp. 3670-3679, 2024, doi: 10.1109/TTE.2023.3315729.
5. I. T. Halidou, **M. H. Elkholy**, T. Senjyu, T. Said, and M. M. Gamil, "Optimal microgrid planning for electricity security in Niamey: A strategic response to sudden supply disruptions from neighboring sources," *Energy Conversion and Management*, vol. 326, p. 119529, 2025/02/15/ 2025, doi: <https://doi.org/10.1016/j.enconman.2025.119529>.
6. Ahmed, Shoaib, Yongyi Huang, Mitsunaga Kinjo, Tomonobu Senjyu, Dongran Song, and **M. H. Elkholy**. "Sustainable and Reliable Energy Management for Urban Hybrid Energy Systems-A Case Study in Islamabad Pakistan on Hydrogen and Battery Integration Using Transient Search Optimization Algorithm." *Results in Engineering* (2025): 106272.
7. Ali, E.S., **Elkholy, M.H.**, Senjyu, T. *et al.* A flexible multi-agent system for managing demand and variability in hybrid energy systems for rural communities. *Sci Rep* **15**, 16255 (2025). <https://doi.org/10.1038/s41598-025-01288-5>
8. **M. H. Elkholy**, Tomonobu Senjyu, Mahmoud M. Gamil, Mohammed Elsayed Lotfy, Dongran Song, Gul Ahmad Ludin, Ahmad Shah Irshad, and Taghreed Said "Implementation of a multistage predictive energy management strategy considering electric vehicles using a novel hybrid optimization technique," *Journal of Cleaner Production*, vol. 476, p. 143765, 2024/10/15/ 2024, doi: <https://doi.org/10.1016/j.jclepro.2024.143765>.
9. **M. H. Elkholy**, Tomonobu Senjyu, Mahmoud Elymany, Mahmoud M. Gamil, M. Talaat, Hasan Masrur, Soichiro Ueda, and Mohammed Elsayed Lotfy "Optimal resilient operation and sustainable power management within an autonomous residential microgrid using African vultures optimization algorithm," *Renewable Energy*, vol. 224, p. 120247, 2024/04/01/ 2024, doi: <https://doi.org/10.1016/j.renene.2024.120247>.
10. Elazim, S.M.A., **Elkholy, M.H.**, Elgarhy, A. *et al.* Enhancing stability and power quality in electric vehicle charging stations powered by hybrid energy sources through harmonic mitigation and load management. *Sci Rep* **15**, 28077 (2025). <https://doi.org/10.1038/s41598-025-14143-4>
11. Song, Dongran, Keli Chen, Runxin Chen, Xinyu Fan, Jian Yang, Mi Dong, M. Talaat, and **M. H. Elkholy**, "A bi-level optimization approach for siting and sizing of distributed wind-storage power systems in rural areas based on MCTS and IPSO algorithms," *Energy Conversion and Management*, vol. 342, p. 120173, 2025/10/15/ 2025, doi: <https://doi.org/10.1016/j.enconman.2025.120173>.
12. **M. H. Elkholy**, M. Elymany, S. Ueda, I. T. Halidou, H. Fedayi, and T. Senjyu, "Maximizing microgrid resilience: A two-stage AI-Enhanced system with an integrated backup system using a novel hybrid optimization algorithm," *Journal of Cleaner Production*, vol. 446, p. 141281, 2024/03/25/ 2024, doi: <https://doi.org/10.1016/j.jclepro.2024.141281>.
13. **M. H. Elkholy**, Taghreed Said, Mahmoud Elymany, Tomonobu Senjyu, Mahmoud M. Gamil, Dongran Song, Soichiro Ueda, and Mohammed Elsayed Lotfy. "Techno-economic configuration of a hybrid backup system within a microgrid considering vehicle-to-grid technology: A case study of a remote area," *Energy Conversion and Management*, vol. 301, p. 118032, 2024/02/01/ 2024, doi: <https://doi.org/10.1016/j.enconman.2023.118032>.
14. **M. H. Elkholy**, Tomonobu Senjyu, Hamid Metwally, M. A. Farahat, Ahmad Shah Irshad, Ashraf M. Hemeida, and Mohammed Elsayed Lotfy. "A resilient and intelligent multi-objective energy management for a hydrogen-battery hybrid energy storage system based on MFO technique," *Renewable Energy*, vol. 222, p. 119768, 2024/02/01/ 2024, doi: <https://doi.org/10.1016/j.renene.2023.119768>.
15. **M. H. Elkholy**, M. Elymany, A. Yona, T. Senjyu, H. Takahashi, and M. Elsayed Lotfy, "Experimental validation of an AI-embedded FPGA-based Real-Time smart energy management system using Multi-Objective Reptile search algorithm and gorilla troops optimizer," *Energy Conversion and Management*, vol. 282, p. 116860, 2023/04/15/ 2023, doi: <https://doi.org/10.1016/j.enconman.2023.116860>.
16. **M. H. Elkholy**, M. Elymany, H. Metwally, M. A. Farahat, T. Senjyu, and M. Elsayed Lotfy, "Design and implementation of a Real-time energy management system for an isolated Microgrid: Experimental validation," *Applied Energy*, vol. 327, p. 120105, 2022/12/01/ 2022, doi: <https://doi.org/10.1016/j.apenergy.2022.120105>.
17. **M. H. Elkholy**, T. Senjyu, M. E. Lotfy, A. Elgarhy, N. S. Ali, and T. S. Gaafar, "Design and Implementation of a Real-Time Smart Home Management System Considering Energy Saving," *Sustainability*, vol. 14, no. 21, doi: 10.3390/su142113840.

18. **M. H. Elkholy**, H. Metwally, M. A. Farahat, M. Nasser, T. Senjyu, and M. E. Lotfy, "Dynamic centralized control and intelligent load management system of a remote residential building with V2H technology," *Journal of Energy Storage*, vol. 52, p. 104839, 2022/08/01/ 2022, doi: <https://doi.org/10.1016/j.est.2022.104839>.
19. **M. H. Elkholy**, H. Metwally, M. A. Farahat, T. Senjyu, and M. Elsayed Lotfy, "Smart centralized energy management system for autonomous microgrid using FPGA," *Applied Energy*, vol. 317, p. 119164, 2022/07/01/ 2022, doi: <https://doi.org/10.1016/j.apenergy.2022.119164>.
20. Irshad, Ahmad Shah, **M. H. Elkholy**, Nahar F. Alshammari, Gul Ahmad Ludin, Tomonobu Senjyu, Gabor Pinter, and Alexey Mikhaylov. "Novel Approach for Energy Balancing with Intermittent Renewable Energy Source Using Multi-Objective Genetic Algorithm." *IEEE Access* (2024). doi: [10.1109/ACCESS.2024.3507216](https://doi.org/10.1109/ACCESS.2024.3507216).
21. L. Pai, T. Senjyu, and **M. H. Elkholy**, "Integrated Home Energy Management with Hybrid Backup Storage and Vehicle-to-Home Systems for Enhanced Resilience, Efficiency, and Energy Independence in Green Buildings," *Applied Sciences*, vol. 14, no. 17, doi: <https://doi.org/10.3390/app14177747>.
22. A. Shah Irshad, Nqibullah Kargar, **M. H. Elkholy**, Gul Ahmad Ludin, Said Elias, Ahmadullah Hilali, Tomonobu Senjyu, Mahmoud M. Gamil, and Gabor Pinter "Techno-economic evaluation and comparison of the optimal PV/Wind and grid hybrid system with horizontal and vertical axis wind turbines," *Energy Conversion and Management: X*, vol. 23, p. 100638, 2024/07/01/ 2024, doi: <https://doi.org/10.1016/j.ecmx.2024.100638>.
23. A. Shah Irshad, Mohammad Naseer Zakir, Sher Shah Rashad, Mohammed Elsayed Lotfy, Alexey Mikhaylov, **M. H. Elkholy**, Gabor Pinter, and Tomonobu Senjyu. "Comparative analyses and optimizations of hybrid biomass and solar energy systems based upon a variety of biomass technologies," *Energy Conversion and Management: X*, vol. 23, p. 100640, 2024/07/01/ 2024, doi: <https://doi.org/10.1016/j.ecmx.2024.100640>.
24. S. Ahmed, Y. Huang, Q. Tayyab, T. senjyu, and **M. H. Elkholy**, "Identification of power grids low-frequency oscillations through a combined MEEMD-Prony method," *Energy Reports*, vol. 11, pp. 4245-4253, 2024/06/01/ 2024, doi: <https://doi.org/10.1016/j.egyr.2024.04.015>.
25. M. Talaat, Ahmed Reda Mohamed, Bishoy E. Sedhom, Ahmed Wahba, **M. H. Elkholy**, Nourhan M. Elbehairy, Dongran Song, and Tomonobu Senjyu. "Monolithic design of self-adaptive CMOS converter and robust event-triggered consensus control for integration of multi-renewable energy sources with battery storage system," *Journal of Energy Storage*, vol. 88, p. 111498, 2024/05/30/ 2024, doi: <https://doi.org/10.1016/j.est.2024.111498>.
26. A. S. Irshad, Soichiro Ueda, Masahiro Furukakoi, Mohammad Naseer Zakir, Gul Ahmad Ludin, **M. H. Elkholy**, Atsushi Yona, Said Elias, and Tomonobu Senjyu "Novel integration and optimization of reliable photovoltaic and biomass integrated system for rural electrification," *Energy Reports*, vol. 11, pp. 4924-4939, 2024/06/01/ 2024, doi: <https://doi.org/10.1016/j.egyr.2024.04.057>.
27. Q. Tayyab, N. A. Qani, **M. H. Elkholy**, S. Ahmed, A. Yona, and T. Senjyu, "Techno-economic configuration of an optimized resident microgrid: A case study for Afghanistan," *Renewable Energy*, vol. 224, p. 120097, 2024/04/01/ 2024, doi: <https://doi.org/10.1016/j.renene.2024.120097>.
28. A. S. Irshad, G. Ahmad Ludin, S. Ludin, **M. H. Elkholy**, S. Elias, and T. Senjyu, "Integration and performance analysis of optimal large-scale hybrid PV and pump hydro storage system based upon floating PV for practical application," *Energy Conversion and Management: X*, vol. 22, p. 100599, 2024/04/01/ 2024, doi: <https://doi.org/10.1016/j.ecmx.2024.100599>.
29. W. Meng, , Dongran Song, Liansheng Huang, Xiaojiao Chen, Jian Yang, Mi Dong, M. Talaat, and **M. H. Elkholy** "Distributed energy management of electric vehicle charging stations based on hierarchical pricing mechanism and aggregate feasible regions," *Energy*, vol. 291, p. 130332, 2024/03/15/ 2024, doi: <https://doi.org/10.1016/j.energy.2024.130332>.
30. D. Song, Xiao Tan, Xiaofei Deng, Jian Yang, Mi Dong, **M. H. Elkholy**, M. Talaat, and Young Hoon Joo. "Rotor equivalent wind speed prediction based on mechanism analysis and residual correction using Lidar measurements," *Energy Conversion and Management*, vol. 292, p. 117385, 2023/09/15/ 2023, doi: <https://doi.org/10.1016/j.enconman.2023.117385>.
31. D. A. Konneh, H. O. Howlader, **M. H. Elkholy**, and T. Senjyu, "An Agile Approach for Adopting Sustainable Energy Solutions with Advanced Computational Techniques," *Energies*, vol. 17, no. 13, doi: <https://doi.org/10.3390/en1713150>.
32. G. A. Ludin, Mohammad Amin Amin, Ahmad Shah Irshad, Soichiro Ueda, Zakirhussain Farhad, **M. H. Elkholy**, and Tomonobu Senjyu. "Power transmission in Afghanistan: Challenges, opportunities and proposals," *AIMS Energy*, vol. 12, no. 4, pp. 840-871, 2024, doi: [10.3934/energy.2024040](https://doi.org/10.3934/energy.2024040).
33. M. Talaat, **M. H. Elkholy**, A. Alblawi, and T. Said, "Artificial intelligence applications for microgrids integration and management of hybrid renewable energy sources," *Artificial Intelligence Review*, vol. 56, no. 9, pp. 10557-10611, 2023/09/01 2023, doi: [10.1007/s10462-023-10410-w](https://doi.org/10.1007/s10462-023-10410-w).
34. I. Tahirou Halidou, H. Or Rashid Howlader, M. M. Gamil, **M. H. Elkholy**, and T. Senjyu, "Optimal Power Scheduling and Techno-Economic Analysis of a Residential Microgrid for a Remotely Located Area: A Case Study for the Sahara Desert of Niger," *Energies*, vol. 16, no. 8, doi: <https://doi.org/10.3390/en16083471>.

35. M. Talaat, A. Alblawi, M. Tayseer, and **M. H. Elkholy**, "FPGA control system technology for integrating the PV/wave/FC hybrid system using ANN optimized by MFO techniques," *Sustainable Cities and Society*, vol. 80, p. 103825, 2022/05/01/ 2022, doi: <https://doi.org/10.1016/j.scs.2022.103825>.
36. M. Talaat, A. Elgarhy, **M. H. Elkholy**, and M. A. Farahat, "Integration of fuel cells into an off-grid hybrid system using wave and solar energy," *International Journal of Electrical Power & Energy Systems*, vol. 130, p. 106939, 2021/09/01/ 2021, doi: <https://doi.org/10.1016/j.ijepes.2021.106939>.
37. Alblawi, T. Said, M. Talaat, and **M. H. Elkholy**, "PV solar power forecasting based on hybrid MFFNN-ALO," in 2022 13th International Conference on Electrical Engineering (ICEENG), 29-31 March 2022 2022, pp. 52-56, doi: 10.1109/ICEENG49683.2022.9782040.
38. M. Dong, Ya Li, Dongran Song, Jian Yang, Mei Su, Xiaofei Deng, Lingxiang Huang, **M. H. Elkholy**, and Young Hoon Joo "Uncertainty and global sensitivity analysis of levelized cost of energy in wind power generation," *Energy Conversion and Management*, vol. 229, p. 113781, 2021/02/01/ 2021, doi: <https://doi.org/10.1016/j.enconman.2020.113781>.
39. M. Talaat, **M. H. Elkholy**, and M. A. Farahat, "Operating reserve investigation for the integration of wave, solar and wind energies," *Energy*, vol. 197, p. 117207, 2020/04/15/ 2020, doi: <https://doi.org/10.1016/j.energy.2020.117207>.
40. A. Alblawi, **M. H. Elkholy**, and M. Talaat, "ANN for Assessment of Energy Consumption of 4 kW PV Modules over a Year Considering the Impacts of Temperature and Irradiance," *Sustainability*, vol. 11, no. 23, doi: <https://doi.org/10.3390/su11236802>.
41. M. Talaat, M. A. Farahat, and **M. H. Elkholy**, "Renewable power integration: Experimental and simulation study to investigate the ability of integrating wave, solar and wind energies," *Energy*, vol. 170, pp. 668-682, 2019/03/01/ 2019, doi: <https://doi.org/10.1016/j.energy.2018.12.171>.
42. **M.H. Elkholy** et al., "Intelligent Dispatch-Based Optimization of Hybrid Renewable Architectures Incorporating Electric Mobility and Hydrogen Storage for Commercial Energy Systems," *IEEE Access*, pp. 1-1, 2026, doi: 10.1109/ACCESS.2026.3661790.
43. Chen, Runxin, Liqing Liao, Dongran Song, Chuxi Wei, Jian Yang, M. Talaat, and **M. H. Elkholy**. "Two-stage optimization scheduling strategy for virtual power plants considering the correlated uncertainty of distributed energy storage in extreme scenarios." *Applied Energy* 409 (2026): 127477.

UNDER REVIEW

44. Dongran Song, Guoyang Shen, Lei Wang, Jian Yang, Mi Dong, **M. H. Elkholy**, M. Talaat, and Young Hoo Joo, "Yaw Optimization for Floating Offshore Wind Farm Based on Monte Carlo Tree Search Algorithm", *IEEE Transactions on Automation Science and Engineering*, **minor review**
45. Ali, E.S., **Elkholy, M.H.**, *et al.* Comparative Analysis and Optimization of Hybrid Renewable Energy Architectures Featuring Electric Mobility and Hydrogen Storage for Commercial Infrastructure Using Intelligent Energy Dispatch Algorithm, *International Journal of Electrical Power and Energy Systems*, **minor review**
46. Javeria Noor, Dongran Song, Asifa Yousaf, Mi Dong, Jian Yang, M. Rizk-Allah, **M. H.Elkholy**, M. Talaat, " Bipolar DC Microgrid Optimization Using Novel SEPIC-Cuk Fused Converter with Fuzzy Logic Based Droop Control", *Energy*, **under review**
47. Nyam Jargalsaikhan, M.Talaat, **M. H. Elkholy**, Tomonobu Senjyu, "Effect of ambient temperature variance on power output of wind turbine and mitigation strategy", *Energy*, **Under review**
48. Qudratullah Tayyab, Abdul Rahman Sorgul, Gul Ahmad Ludin, Shoaib Ahmed, **M. H. Elkholy**, Harun Or Rashid Howlader, Said Elias and Tomonobu Senjyu, "Empowering Smart Grids: Maximizing Fuel Savings and Grid Stability through Optimal Energy Storage and Unit Commitment", *Energy Conversion and Management: X*, **Accepted**
49. **M. H. Elkholy**, Artificial Intelligence-Based Joint Planning and Operational Optimization of Electric Vehicle Charging Stations and Distributed Generators for Enhanced Reliability and Resilience in Power Distribution Systems, *IEEE Transactions on Automation Science and Engineering*, **Under Review**
50. **M. H. Elkholy**, and Elgarhy, A., High-Fidelity Simulation and Transient Stability Assessment of Power Quality and Damping Behavior in Multi-Source Electric Vehicle Charging Stations with Combined Load Configurations Operating, *eTransportation*, **Under Review**
51. **M. H. Elkholy**, Optimal Design and Energy Management of a Renewable-Powered Microgrid for EV Charging Stations with Park-and-Ride Integration: A Case Study in Liaoning Province, China, *Protection and Control of Modern Power Systems*, **Under Review**
52. **M. H. Elkholy**, Synergistic E-Mobility-Aware Power Scheduling for Park-and-Ride Electric Vehicle Charging Infrastructure Incorporating Bidirectional Vehicle-to-Station Transfer on Okinawa Island Using a Hippopotamus Optimization Algorithm, *Applied Energy*, **Under Review**

54. **M. H. Elkholy**, Integrated Spatial and Capacity Optimization of Renewable-Powered Park-and-Ride Electric Vehicle Charging Hubs for Urban Mobility Enhancement Congestion Mitigation and Power System Stability, **eTransportation**, **Under Review**